

DSCI-D590-9369 Final Project Report:

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Introduction:

It's no secret that extreme weather events have become more commonplace with our ever changing climate. While the frequency of extreme snowstorms in the US East coast has seemingly increased over time, I've honestly felt the opposite over here in the Pacific Northwest, or more specifically Washington state. And while I'm not jealous of the severe hindrances & disruptions presented by extreme snowstorms on my daily schedule, I am curious to see how trends in snow have changed over time in my home state of Washington state. Due to my background with a Bachelors of Science in climate, I find this topic very interesting as changes in yearly weather could impact snowpack density which in turn, impacts our access to water.

Therefore, for my project, I seek to answer: How have extreme snow events and overall snow cover changed in Washington within the 21st century? More specifically, my hypothesis before literature review will be:

Both extreme snow weather events and overall snow cover have decreased in Washington state over the 21st century.

I hope to resolve my hypothesis by utilizing regression and geospatial analysis to identify anomalies in extensive weather datasets. I will most likely need to take a highly specific subset of my dataset because most weather datasets, like from NOAA, are massive. Also, by comparing

those anomalies to overall seasonal trends, using concepts we've mastered from class, I hope to meaningfully compare these anomalies to average trends in weather from previous years.

Literature Review:

Due to my background in climate science for my Bachelors, I felt like this would be a perfect blend of this program and my previous studies. Therefore in terms of research, I first prioritized resources that tracked snow related phenomenon over time. The most useful resource I found was a relatively recent paper about a time series construction of snowfall observations in both Washington and Oregon (Christy & Redmond, 2022). This resource is incredibly useful because it provides both a framework for my own analysis and squeaky clean data allowing me to derive conclusions about the seasonality and anomalies from the data. Likewise, I found another incredibly relevant paper about tracking "snow drought" in the Pacific Northwest (Roberts-Pierel & Livneh 2024). However, I was wary when referring to this resource because it already supports my conclusion from part one that extreme snow weather events and overall snow cover have decreased in Washington state. These resources cover overall snowfall cover through the entire state, but I knew to have the strongest and most unbiased analysis possible, I should reference data from other relevant sources and phenomenon related to snow.

From my Bachelors degree, I know that snowpack volume (essentially the ice at the top of mountains that melts in the spring to form streams & rivers) is crucial in determining water security. However, trends in the snowpack volume over time are relevant to this study because the volume of the snowpack is simply a function of added falling snow in the winter subtracted by melted snow in the spring/summer. Therefore, through papers about snowpack volume, I

cross referenced information from the previous papers and learned more about snow fall trends in mountainous regions. Specifically, I looked at a paper about whether the snowpack in the Washington Cascades has declined (Mote & Hamlet, 2008) and a paper about the overall snowpack trends in the Cascades (Stoelinga, 2009). Funnily enough, I actually read the Stoelinga paper as part of my undergraduate studies. Furthermore, I looked at the U.S. Department of Agriculture's paper on snow water equivalent in the Pacific Northwest (2022) to further support the other papers. While these papers are good at identifying the overall trend in snowfall, I still needed data to identify trends in anomalous extreme snow events as it's a core component of this study.

Thus, for the final resources in this literature review, I wanted them to focus on studies that explored anomalous behavior in snowfall. I found a great paper about spatiotemporal variability of snowfall in the Pacific Northwest (Catalano & Loikith, 2019). I also found a more narrow resource about a specific extreme snow event in 2012 (Bumbaco, 2017), but it still provided general information about extreme events to coincide with Catalano's paper.

Through reviewing the literature I found for this project, I seek to refine my research questions and gain a general understanding of how I should practically proceed with the project. Furthermore, by viewing snow phenomena from a diverse set of sources, I hope to eliminate potential bias and bolster my findings.

For the purpose of this review, I decided to split the sources into three sections. This first section being solely focused on overfall snowfall cover. As previously stated, the most important

paper for this project was Christy et. al's (2022) paper on the time series construction of snowfall observation. In terms of data, the paper encapsulates measurements from 500 different weather stations across Oregon and Washington. However, the period of the time series was a little more complex as the researchers ran into multiple complications in aggregating the data. The paper included measurements from 1879 to 2020, but data prior to 1940 was not easily accessible in their archive and their archive wasn't completely accurate with some data incorrectly being flagged as missing or eliminated reports due to digitalization. Despite these complications, I felt the work done in this paper was still relevant to my study because I'm primarily concerned with the trends from the late 20th century and the 21st century. Therefore, I'm going to proceed with a time range of around 1960 to the 2020s for my project. The paper aimed to study the changes in snow in geographically similar sub-regions with Washington state consisting of 9 different sub-regions. From there, a time series was constructed for every station in a subregion to the sub-regions reference station where the "seasonal median snowfall" (Christy et. al, 2022) could then be compared. Ultimately, for these 9 sub-regions, the paper found a very slight negative trend in snowcover in 1890 to 2020 in areas of low altitude with less change in areas of higher altitude (like snowpacks). This finding contrasts with my previous expectations that there'd be a noticeable change in snow cover over time, but as this paper had a larger scope, it was less concerned with yearly snow anomalies. Thus, there could still be overall trends in the frequency, severity, or duration of extreme snow events that could be highlighted in the other papers.

Continuing the first section, the next source was Roberts-Pierel & Livneh's (2024) paper on snow drought in the Pacific Northwest. While this paper has an even larger geospatial scope

than my project (including Idaho and Oregon), I appreciate the recentness and the slightly different approach to quantifying snow from Christy's paper. Data used in this paper was sourced from both direct measurements and optical remote sensing with some additional gridded modeling. Roberts-Pierel defines snow drought as deficits in snowfall and it can be used to "identify intra-annual differences in seasonal snow patterns and decadal trends in seasonal snow change" (Roberts-Pierel & Livneh 2024). The primary objectives of the paper were to identify the frequency and common locations of snow drought while relying on the optical remote sensing approach first. However, they later found that optical sensing struggled at detecting moderate cases of snow drought. Compared to the previous paper, this study encompassed data from 3 geospatial datasets over a smaller time period of 1991-2020. While this time period is smaller than ideal for my project, the data from this time range is more accurate than the previous source because Roberts-Piere & colleagues had access to cleaner aggregated data. Their specific scientific definition of a snow drought was derived from a different paper by Dierauer et al (they changed how Snow Water Equivalent was calculated), but essentially a severe snow drought is defined by the precipitation, temperature, and SWE conditions being lower than the average. Ultimately, Roberts-Piere's study found that the "net snow drought occurrences increased over the last three decades" (Roberts-Pierel & Livneh 2024) by roughly 10 to 15%, but this "was not a statistically significant result based on annual data" (Roberts-Pierel & Livneh 2024). This finding largely coincides with the findings from the Christy paper, but the specificity of increasing snow droughts over the last 3 decades could point to the behavior we will be observing in the future. However, they also found that the droughts varied geospatially with the eastern Pacific Northwest (or in this case not Washington) observing that **"warm/dry snow drought rarely occurred in ≥ 10 years in the eastern portion of the**

study area” (Roberts-Pierel & Livneh 2024). Overall, I may need to lower my expectations of how pronounced the trends in snow will be for my project or choose to focus on a more specific phenomenon in Washington's snow-water system.

For the second section of this review, I will be focusing on changes in snowpack volume and cover. The most important snowpack for Washington state is found in the Cascade mountain range and the first paper by Mote & Hamlet (2008) takes a look at whether it has declined in recent history. Data from this paper was almost exclusively sourced from direct measurements with over 1000 locations sampled and measured roughly monthly. They used VIC simulations (Variable Infiltration Capacity which is commonly used for hydrological systems) to model and predict both the balance of water and the snowpack. However, the accuracy of a VIC simulation is variable because “there are several reasons why the SWE simulated by VIC and the SWE measured at a snow course might differ: terrain, temporal offsets, interpolation of driving data, spurious trends in driving data, and land-cover change” (Mote & Hamlet, 2008). Despite this, the model was still good at predicting “topographic gradients in snowpack trends” (Mote & Hamlet, 2008) and overall is still relevant for analysis. Given both the model & direct measurements, the researchers found that the size of the snowpack by the start of every annual spring has declined by 15-35% compared to springs from the mid 20th century. Overall, this coincides with the negative trends in snow from the other papers, but these findings appear more significant than from the other papers. However, it should be acknowledged that this paper is less recent than the Christy and Roberts-Pierel papers, so there could be more modern trends that contradict findings in Mote’s paper. The variance of 20% is also pretty large, but the minimum bound of 15% is still

statistically significant and is admittedly more consistent with the more moderate changes found from the previous papers.

Supporting these findings, a similar paper by Stoelinga (2009) further studies changes in the Cascade snowpack from a different perspective. Stoelinga aims to measure changes in the Cascadian snowpack over a longer period of time between 1930 and 2009, but direct measurements of the snowpack were not taken until 1950, so some estimation was required. As for their approach, they took an interesting direction by sourcing data from past streamflow (which is directly tied to the amount of melted snow), precipitation, temperature, and direct measurements. However, in terms of meaningfully quantifying measurement, Stoelinga compared all analysis to the 1961-1990 mean. With this method Stoelinga found that “over the entire period of record (1930-2007) is -23% of the 1961-1990 normal, although this trend does not quite meet the 95% confidence level for a non-zero trend” (Stoelinga, 2009). The seasonality of the snowpack also shifted by 5 days, but that’s not enough to be statistically significant. Similar to Mote’s paper, we see a negative trend that falls between 15 - 35% and Stoelinga acknowledges that the data had not reached 95% confidence which also coincides with the variability & uncertainty from the Mote paper. From both of these sources, I now believe it’s pretty hard to establish trends when it comes to snowpacks because of natural variability in the system which means I should consider its priority when it comes to my own project. Both papers are rather outdated so I wanted a more recent resource to support these papers.

The Snow Water Equivalent (SWE) paper from the Department of Agriculture (2022) provided current context for both the snowcover and snowpack sections of this literature review.

They provided a figure mapping changes in peak snowpack volume over the entire Pacific Northwest and they found a more pronounced change in the snowpack seasonality compared to the Stolinga paper with peak volume occurring 20 days earlier in the year. The shift likely due to the modernity of this report. Despite a slight downwards trend, the Department of Agriculture's report also found the timing and total peak volume of each year is becoming more variable and erratic. This could indicate an increasing lack of stability, but the paper doesn't go into much detail about this phenomenon.

For the third & final section of this review, I aim to focus on snow anomalies and how their frequency has changed over time. I found an incredible paper on this topic by Catalano & colleagues (2019) that studies the spatiotemporal variability of snowfall in the present and in the future with site specificity. Like many other papers, the data consisted of observations of snow water equivalent and precipitation from the Snowpack Telemetry Network (automated sensors which eliminate the need for human-powered measurements). Given the data, the researchers then constructed models that would project snowfall frequency in accordance with different global CO₂ emissions scenarios. The researchers defined a wet day as a day with more than 7mm of precipitation. From the network, the researchers found that stations in the Cascade mountain range over 1981 - 2005 "were experiencing less than 50% of wet days as snow" and "over two thirds of stations across Oregon and Washington receive snowfall for less than one third of average cold-season wet days" (Catalano & Loikith, 2019). Which means the majority of snowfall is being concentrated to $\frac{1}{3}$ of the cold-season. However, as the researchers acknowledge, this approach of measuring rain vs snow is impacted by overall trends in weather & rain which is also impacted by climate change. The study also found that the frequency of

snowfall was the lowest in areas of low elevation which supports my previous hypotheses. However, as previous papers have agreed upon, Catalano agrees that overall changes over the last 50 years have been minimal, but Catalano and colleagues have found worrying trends in modern history that indicate subsequent frequency loss in the future. In accordance with different warming scenarios, the researchers created multiple models that predict the frequency of snowfall by the end of the 21st century with some predicting as much as a 17% decrease in snowfall frequency. Overall, this resource indicates that snowfall frequency will decrease in the future, but the severity of each snowfall event is still variable.

Speaking of a specific severe snowfall event, I looked at a paper on an avalanche on Snoqualmie Pass (2017) caused by a severe rain-on-snow event. An avalanche was caused on the Snoqualmie pass due to a significant snowfall season in the 2008-2009 winter season preceded by an intense rainstorm causing rapid snow melt. While this resource provided an example of an extreme snow event, I am more concerned with focusing on snowfall whereas this paper primarily focused on the mechanics behind the avalanche. The following paper about an extreme snow/ice event in 2012 by Bumbaco (2012) was more relevant for my project. Bumbaco goes into detail about the conditions leading up to the abnormal snowfall with the “snow and ice storm of January 2012 was one of the highest impact weather events for western Washington in the last few years” (Bumbaco, 2012). This resource appears to contrast with Catalano’s findings, but I’d argue it actually supports the findings, because Catalano’s paper only argues that the frequency of these events are decreasing, not the severity and acknowledges that there is still significant variability in the system.

In conclusion, completing this literature review taught me three key lessons about snowfall in Washington state. Firstly, negative cumulative trends over a 50 year time period of snowfall will be less pronounced than I originally believed. Secondly, using snowpack data to analyze trends in snowpack volume was difficult, variable, and often did not produce statistically significant discoveries. Thirdly, the frequency of snowfall is predicted to decrease, but there are still questions about severity when it does snow. Therefore, for my project, I am going to pivot away from tracking cumulative yearly snowfall and lean into tracking the frequency of snowfall spatially. Furthermore, I will analyze the severity of these snowfall events to examine the correlation between the frequency and intensity of the snowfall events. I seek to answer two questions. How has the frequency vs the intensity of snow storms changed over time? And how will it change in the future due to climate change? After literature review, here is my new hypothesis:

I predict the frequency of snowfall in Washington state from 1985 to 2025 to decrease slightly while the intensity of extreme snow events will increase unproportionately.

Methodology:

For the source of this project's dataset, there is a clear frontrunner in the National Oceanic Atmospheric Administration (otherwise known as NOAA for short). This is a reputable government organization that tracks daily, monthly, and yearly weather phenomena. Actually, most data used in local weather broadcasting is sourced directly from NOAA because all of the data they collect is public. They have an extensive database with daily weather measurements for all 50 states with some of the data even dating all the way back to 1895. Obviously, as seen in the

literature review, the older measurements are more prone to errors because they've been transferred between different recording systems over time. However, for the purpose of this project, I'll be looking at trends over the past 40 years because the measurements are more accurate and anything longer would be an excessive amount of data. Through the many programs I used in this project, with a 40 year time span, I was barely able to subset the data down to 11 million rows and have it function properly in Google Colab.

The first challenge of the project was actually indexing the required data from NOAA. To get the degree of granularity that I wanted, I chose to use the Global Historical Climatology Network's (otherwise known as the GHCN) daily dataset which is massive because it contains measurements for every day of the year. At first, I tried to download the data directly from the website and I was able to subset to Washington State, but it didn't provide the specificity required for this project because the provided datasets included irrelevant columns for my analysis like wind speed, rainfall, and cloud cover. For my project, I only focused on minimum temperature, maximum temperature, snow depth (known as SNOW in the notebook), snow water equivalent, and any additional irrelevant column significantly would've increased the file size. Therefore, the best method I found to retrieve the data was to use Command Prompt on my local computer to index the data from a public NOAA AWS repo which was much faster. Using Command Prompt, I retrieved a separate dataset for each individual year between 1985 and 2025 transported into a singular folder. These datasets were then combined together with a join to only include data from Washington state and contain relevant measurements in the DATA_VALUE column. Ultimately, I was able to reduce the data down to 4 columns and 11 million rows. The columns were ID which refers to the id of the atmospheric station the measurement was taken at,

DATE which is just a timestamp, and ELEMENT which refers to what measurement is in the DATA_VALUE column. The way the dataset is formatted reduces the number of columns by having different measurements like snow depth and minimum temperature be split into separate rows indicated by ELEMENT. I should also add that min and max temperature represent the min and max temperature recorded on that specific day. Being formatted this way does increase the total number of rows, but that's how the AWS data was formatted and not all stations take the same measurements.

Once I was finished subsetting, I uploaded the new dataset called: "wa_snow_temp_daily.csv" to Google Colab to begin cleaning the data. First, using pandas, I casted DATE from object to datetime for future time series analysis and DATA_VALUE from integer to numeric to allow decimals. I also casted both ID and ELEMENT to be categorical to significantly reduce the size of the dataset as both columns have a set range of values they can be. Second, I derived additional columns from existing columns like year and month from DATE. To specifically look at snow trends in just the winter, I created a boolean column that would be true if the month was December, January, or February. Third, I had to convert the temperature measurements in DATA_VALUE as they were technically measured in tenths of a Celsius degree, so I just divided all temperature measurements by 10. After this step, I checked the maximum and minimum values for each category in DATA_VALUE and I realised some of the values were physically impossible like the max temperature recorded being 3143.3 degrees Celsius. However, I realized I shouldn't just blindly remove outliers as a primary goal of my project is to identify trends in extreme weather events. The 3143.4 degrees measurement is clearly a clerical/recording error, so after some NOAA research I found thresholds that are

physically impossible for Washington State. I then subsetting the dataset to only exclude rows that contained measurements outside the thresholds. Finally, I did a quick check for missingness which yielded no missing values and a check for multicollinearity with the four ELEMENT categories which (despite a clear logical correlation between min and max temperature) proved the categories were not directly related to each other.

Ultimately, to thoroughly test my hypothesis, I must analyze differences in temporal trends between extreme snow events and normal snow events within my cleaned dataset. Therefore, I had to define what qualifies as an extreme snow event or storm. After some research in the NOAA documentation and the Catalano & colleagues paper (2019), I found that a severe snow storm is best defined as snowfall that falls between the 90th to 100th percentile of all measurements in a specific state. Given that threshold, I planned to split the dataset into two different regression analyses. The first regression was exclusively for normal snow days or anything below the 90th percentile. The second regression was exclusively for extreme snow days where measurements fall past the 90th percentile. Before the regressions I aggregated snow & temperature into two new dataframes that represented the yearly average snowfall for each station. However, since 9 months out of the year rarely have snow, I did an additional two aggregations for yearly average snow exclusively in the winter months. I took all four of these aggregated dataframes and combined them into a new dataframe called `yearly_stats` which I then verified to ensure the merges worked properly. Then I performed similar aggregations for the normal and extreme snow days for the sake of the corresponding regression models. Both models produced a p-value that was below 0.05 which indicates that they are statistically significant, but I'll go more in depth into each model's performance and insights in the results section.

For further research, I wanted a geospatial analysis on how extreme weather frequency has changed over the last 40 years relative to each station. Therefore, the easiest way to visually communicate this phenomena was to plot each station in terms of latitude and longitude over a rudimentary map of Washington state by the frequency of extreme snow events. From NOAA, I had to download a text file that contained the latitude and longitude of every station as my primary dataframe doesn't contain that information. I joined this text file with the yearly_extremes dataframe (from the extreme snow regression) with respect to ID which essentially added the latitude and longitude columns. Then from a public github repository called PublicaMundi, I downloaded geographic data to set up the background of the plot as Washington state. All of this combined to create a plot that maps out the frequency of extreme snow storms compared to other stations by the darkness of red.

Lastly, I created three figures that provide additional context in the results section of this report. The first plot tracks how the whole year snowfall rate has changed over the past 40 years with a line chart. The second plot shows how snow day frequency is directly correlated to the annual mean minimum temperature which was more of a sanity check rather than a relevant insight. The third and final figure is a geospatial figure that shows how the snow day frequency has changed from 1985 to 2025 for each individual station.

Results:

With the two regression models in place, I compared the results to my hypothesis that the overall trend in snow has decreased while extreme snow storms have increased since 1985. The normal snow regression performed slightly better than the extreme model with a R-squared value

of 0.459 explaining 45.9% of the variance in the model. The results were proven to be statistically significant with a F-statistic of 33.03, and a P-value (otherwise known as the Prob f-statistic) of 1.16e-06. The scatter plot for the frequency of normal snow can be as seen below:

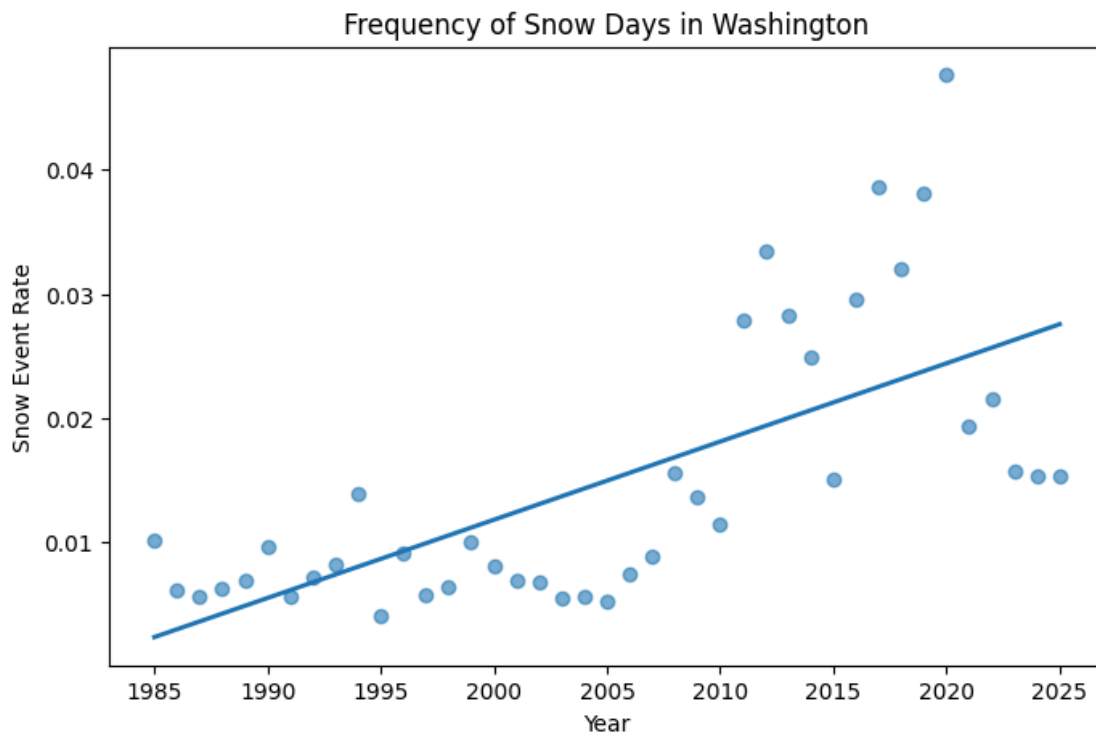


Figure 1: Frequency of Normal Snow Day per Year (1985–2025)

Note. Data from National Centers for Environmental Information (NOAA), Global Historical Climatology Network–Daily (<https://www.ncdc.noaa.gov>).

From the model statistics, the coefficient of YEAR is positive which indicates that normal snow events have become slightly more frequent. This coefficient represents an average 0.06% increase in extreme snow events each year and if we multiply by 40 for the 40 year time span, we get 2.4%. The percentage increase might not seem like much, but it is very significant in terms of climatology as change occurs rather slowly. This challenges my hypothesis because I predicted the overall frequency of snow days to decrease and I’ve calculated a 2.4% increase. That being said, the extreme snow regression produced different results. The normal snow regression

performed slightly worse than the normal model with a R-squared value of 0.360 explaining 36% of the variance in the model. The results were also slightly less statistically significant with a F-statistic of 21.96, and a P-value of 3.36e-05. The scatterplot for the extreme snow regression can be as seen below:

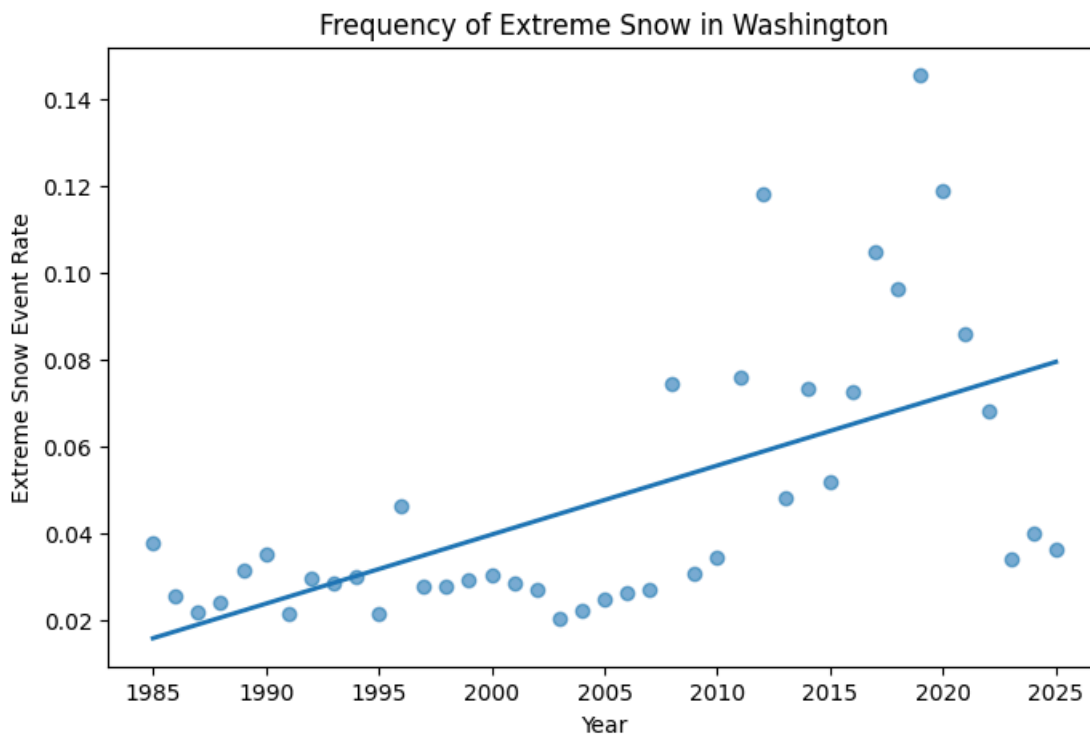


Figure 2: Frequency of Extreme Snow Day per Year (1985–2025)

Note. Data from National Centers for Environmental Information (NOAA), Global Historical Climatology Network–Daily (<https://www.nci.noaa.gov>).

Again, the coefficient of YEAR is positive which indicates that extreme snow events have become more frequent. However, the coefficient represents an average 0.16% increase in extreme snow events each year and if we multiply by 40 for the 40 year time span, we get 6.4%. Therefore, the frequency of extreme snow events has increased faster and unproportionally compared to normal snow behavior which confirms the second half of my hypothesis. Visually both regression plots look very similar with higher snow rates towards the later half of the years

which makes sense because years with higher overall snow would also have a higher likelihood of an extreme snow storm.

To confirm the accuracy of both regressions, I plotted the yearly mean snow day rate for the entire year instead of just for the winter months. Snow doesn't always occur exactly within the three defined winter months, so ensuring overall trend in the yearly mean snow day rate matches the overall trends from the regressions is important for accuracy.

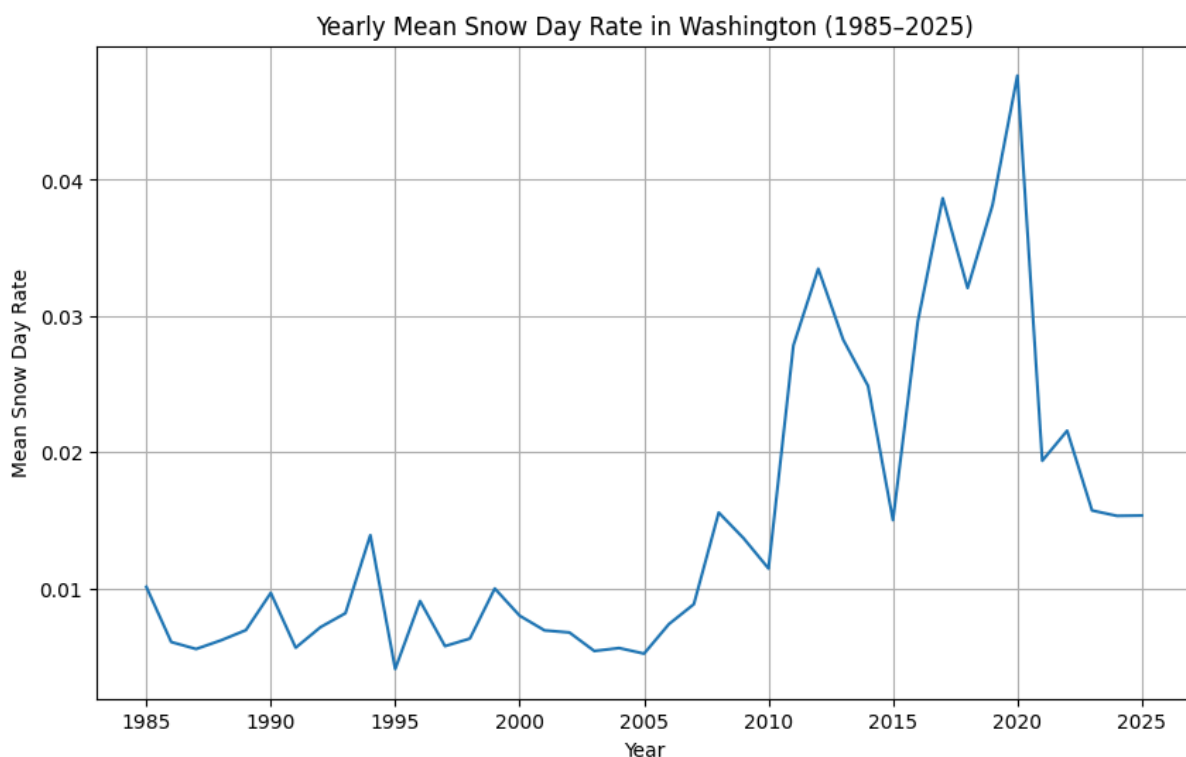


Figure 3: Overall Yearly Mean Snow Date Rate (1985–2025)

Note. Data from National Centers for Environmental Information (NOAA), Global Historical Climatology Network–Daily (<https://www.nci.noaa.gov>).

The line closely resembles the trends in both regression plots with higher measurements occurring towards the end of the year. This plot confirms that the other months excluded from

winter did not have a significant impact on the snow day rate. Therefore, I feel more confident in the accuracy of the regressions or more specifically the normal snow day regression.

With my hypothesis so far half confirmed, it made me wonder if my perception of the lack of snow was a result of where I lived geographically and if other parts of Washington were receiving more snow. Thus, I tested this inference by mapping the extreme snow event occurrence across a simplified map of Washington state by station as seen below:

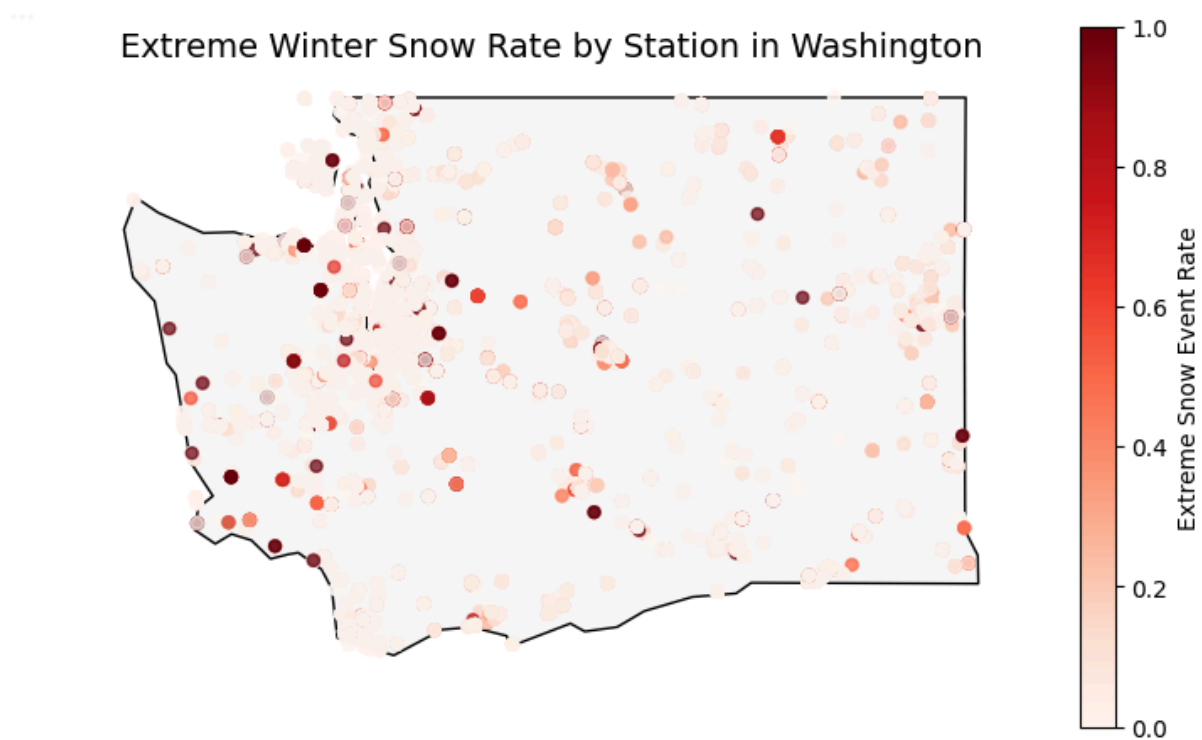


Figure 4: Extreme Winter Snow by Station to Station (1985–2025)

Note. Data from National Centers for Environmental Information (NOAA), Global Historical Climatology Network–Daily (<https://www.nci.noaa.gov>).

Darker red means that the station is experiencing more extreme snow occurrences than compared to other stations. Visually, it appears clear that the stations experiencing the most extreme snow are primarily located towards the west coast compared to central and eastern Washington with

lighter measurements. Geographically this makes sense because the Cascade mountain range separates these two areas which impacts the winds direction, speed, and shear. I live on the Puget Sound which is located towards the top left of the map. While there are multiple white measurements on the shore of the Sound there are multiple red measurements in the nearby area which makes my inference inconclusive. However, since the primary goal of this project is to identify trends over time, I felt that it was important to see how the snow day frequency has changed station to station as seen below:

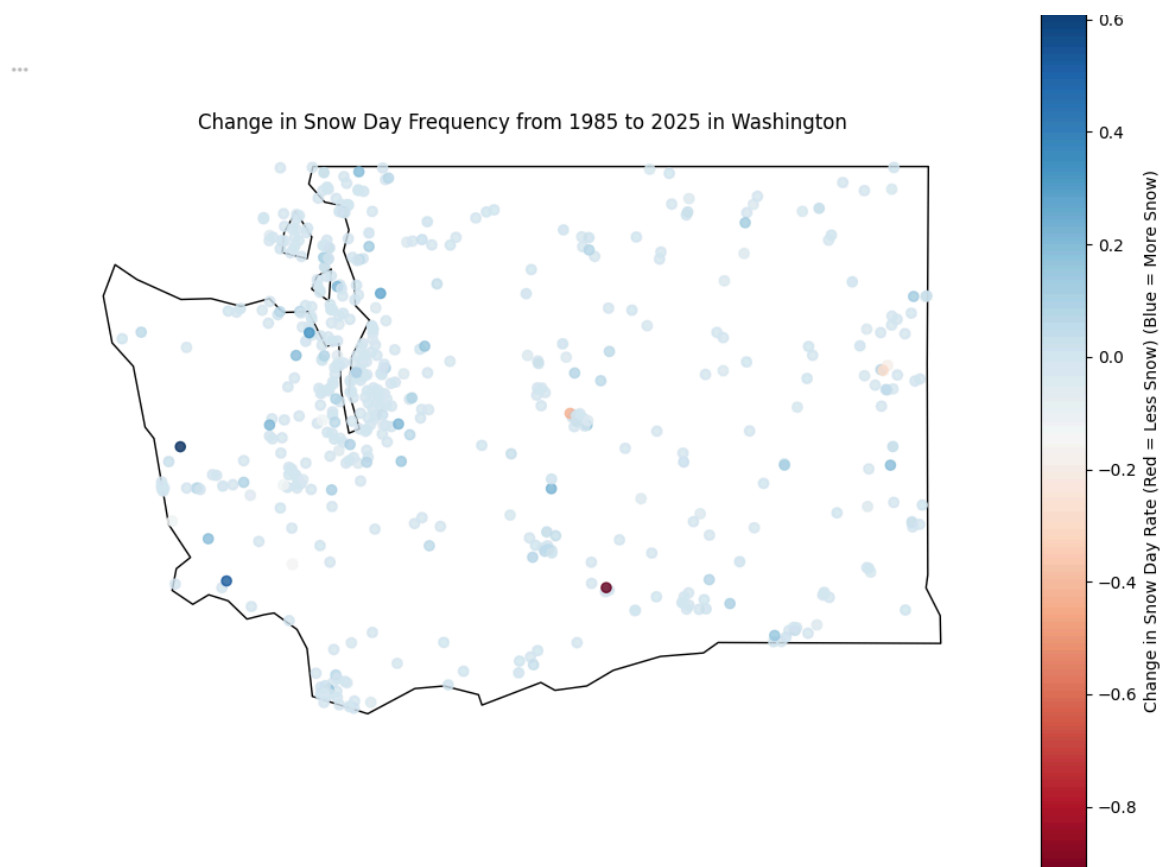


Figure 5: Change in Snow Day Frequency by Station to Station (1985–2025)

Note. Data from National Centers for Environmental Information (NOAA), Global Historical Climatology Network–Daily (<https://www.ncei.noaa.gov>).

Figure 5 compares the frequency of snow in the first half of the time span (1985 - 2005) to the second half of the time span (2005 - 2025). Blue means snow days became more frequent and

red means less frequent. Figure 5 further supports the findings from figure 4 where the deeper blues exist closer to the west coast. Overall, every station is generally blue supporting the results from the regressions completed previously. There were only three stations in the red which can be described through outliers or random noise. Through these geospatial plots, I conclude that western Washington experiences more extreme snow events and has experienced a greater change in the frequency over time.

Conclusions:

To confirm my second hypothesis, I had to prove that the overall frequency of snow days had decreased while the frequency of extreme snowfall had increased over time unproportionately. From my results, I only confirmed that the incident rate of extreme snowfall was increasing faster than the normal incident rate. Overall, according to 1985 to 2025, snow is becoming more common in Washington state rejecting the first half of my hypothesis. Given these results, there are admittedly some limitations with my project and workflow.

My findings are contrary to many of the sources I reviewed in the literature review section. Although the scope of my project was more specialized (just Washington instead of Pacific Northwest) compared to other sources like Catalano and Christy, my experiment was limited in the time span I was covering. While I'm confident that these results are reflective of the last 40 years, these results could be less significant in the context of a 100 years or 150 years as seen in other more professional experiments. However, adding more years to my workflow made indexing/cleaning the data on the programs we've learned about from class more difficult as just 40 years already made the dataset over 11 million rows. Therefore, I acknowledge that the

smaller time span of 40 years is an obvious limitation of my analysis. I should also clarify that the existence of global warming or climate change doesn't disprove this finding. Higher yearly average global temperatures increase the volume of water in the atmosphere through evaporation which is confirmed to increase the severity of water/precipitation related phenomena like hurricanes, atmospheric rivers, and theoretically snowfall. The struggle between the new prevalence of water in the atmosphere versus the higher average winter temperature is currently being studied by experts, but again, a 40 year time span is likely too small to accurately reflect long term trends.

For future directions the project could be taken, I could address the primary limitation by researching a larger time span to address possible interannual variability. I could also explore how the trend in rainfall compares to the trend in snowfall, how much water is added to the ecosystem from snow versus rain, and how future projections of climate warming impact snow day frequency. Building on that last idea, I could construct an experiment that predicts snow frequency in accordance with different Intergovernmental Panel on Climate Change (IPCC) warming scenarios.

Through completing this project, I found that snow has generally become more frequent within the context of the last 40 years, but the frequency of extreme snow storms within the 90th percentile has increased at a faster rate. Therefore, although I must reject my hypothesis, I was ultimately right that extreme snow events are becoming more commonplace which could be explained by the prevalence of evaporated water in the atmosphere.

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